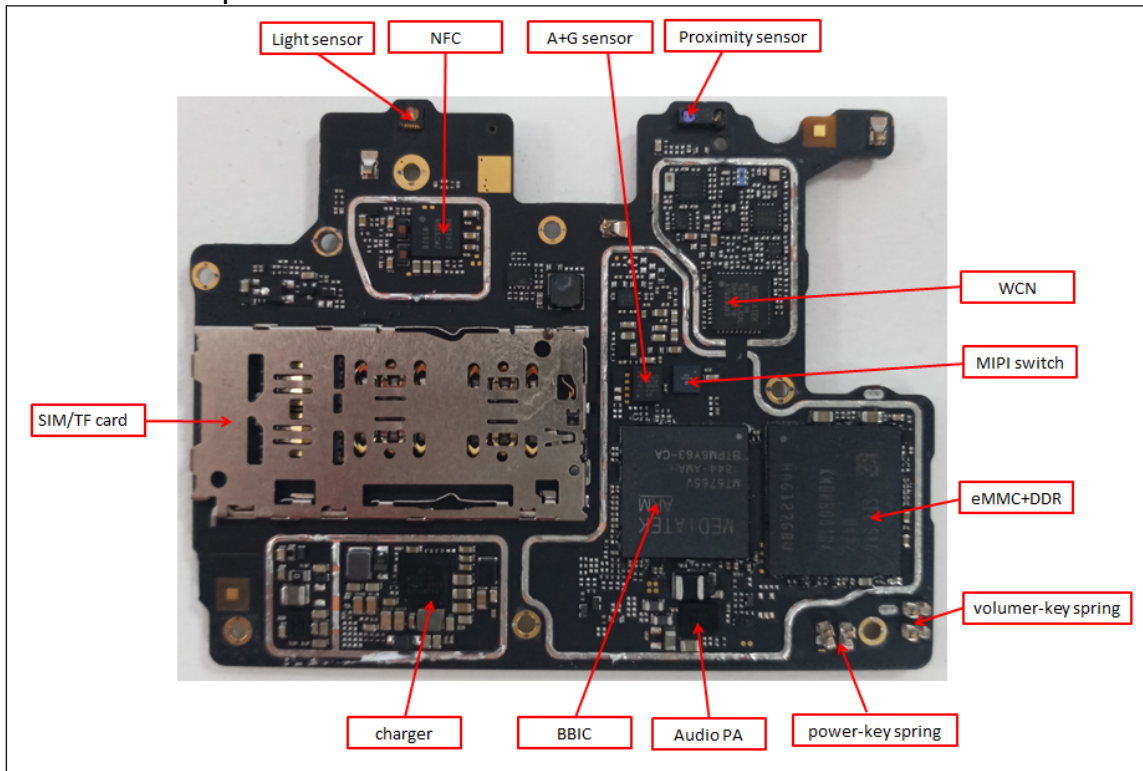


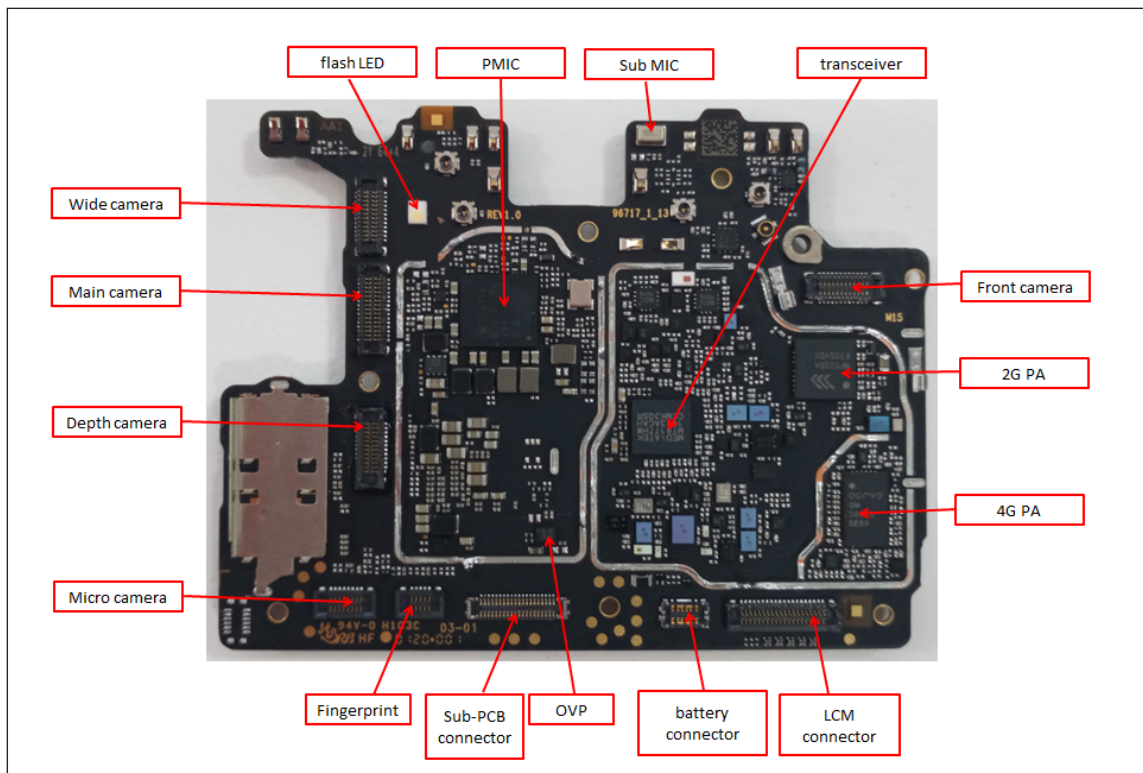
8. Level 3 Repair

8-1. Components Layout

Main board top view

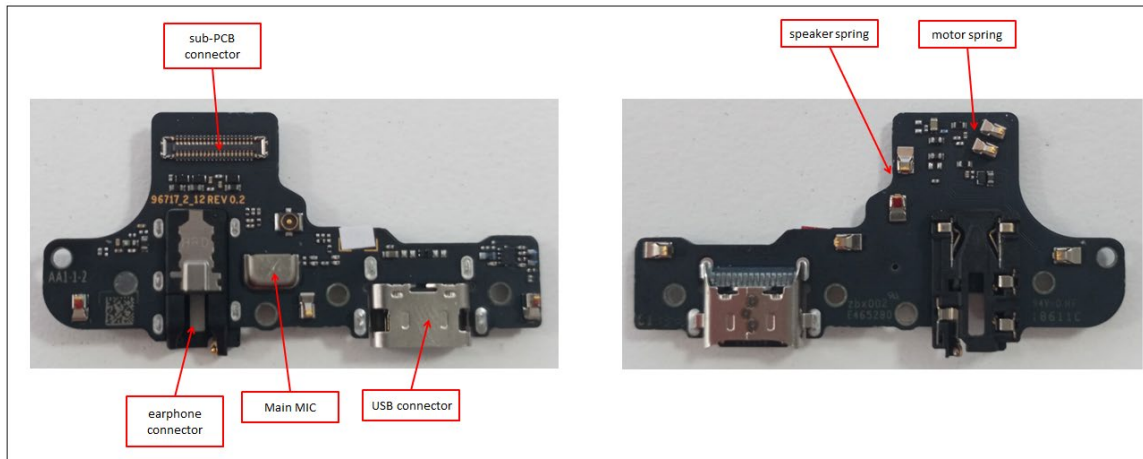


Main board bot view

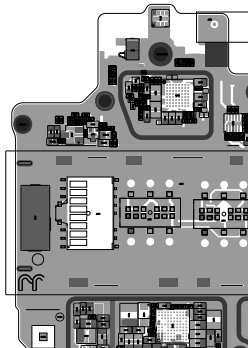


8. Level 3 Repair

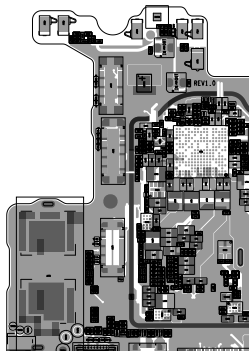
Sub board

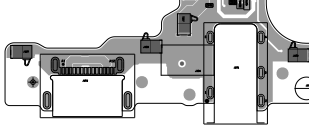


Reference map - Main PBA TOP VIEW



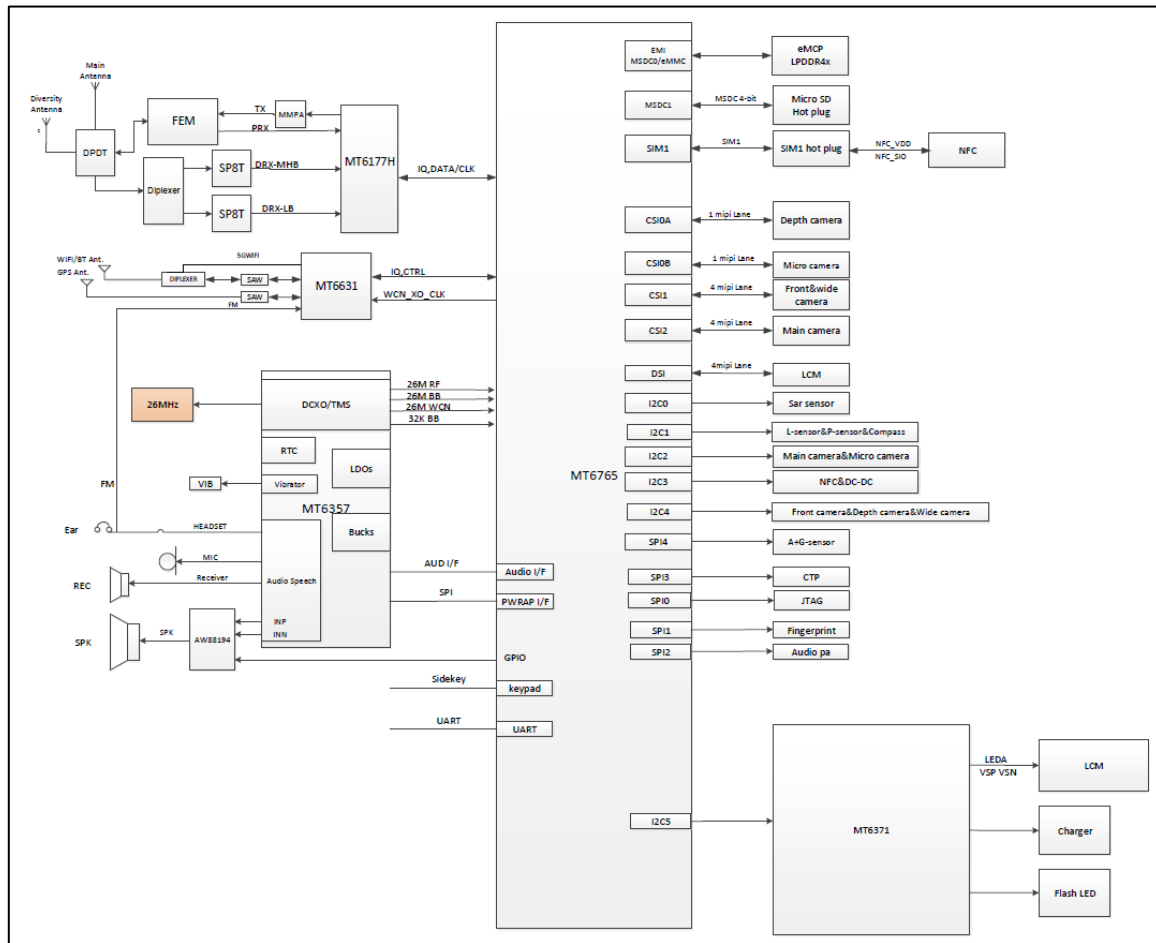
Reference map - Main PBA BTM VIEW



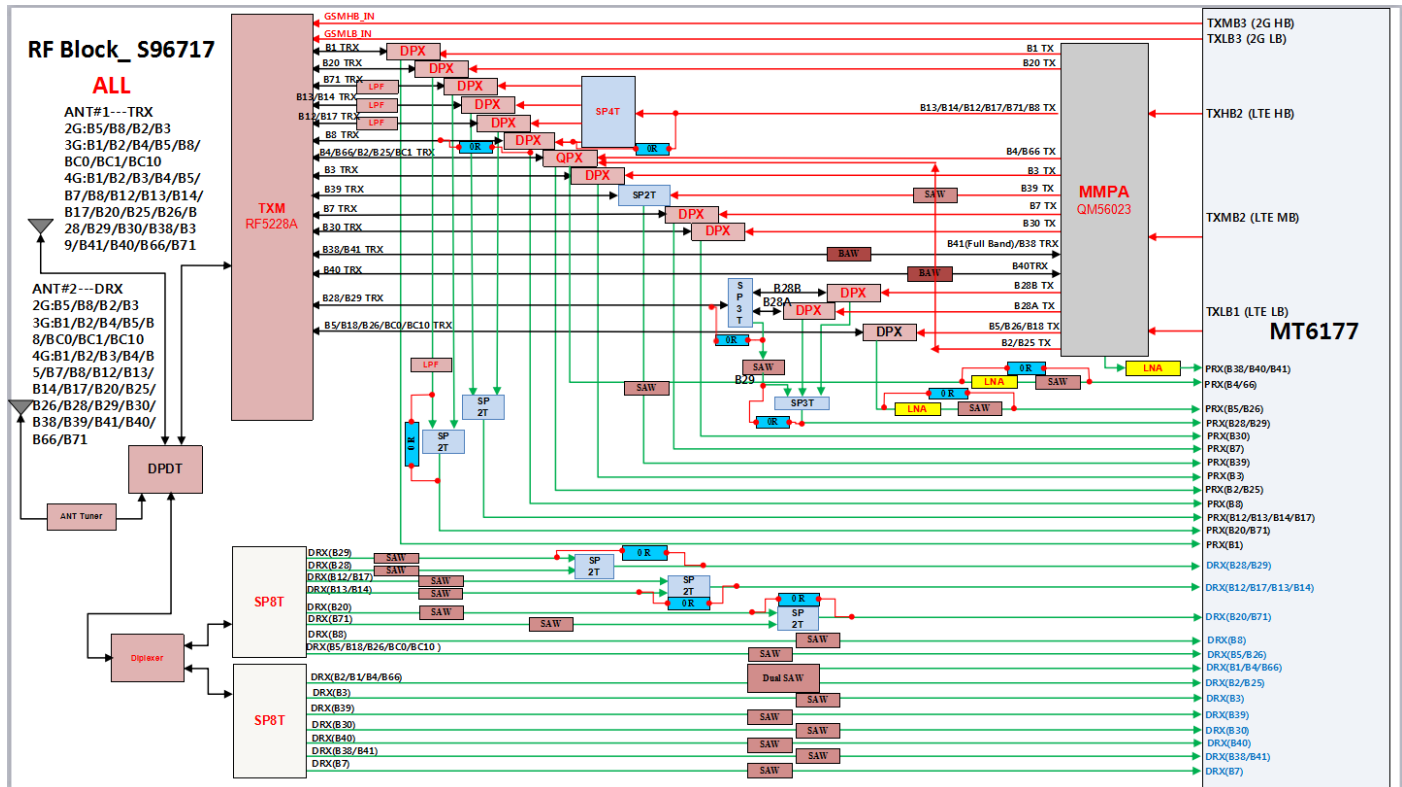




8. Level 3 Repair

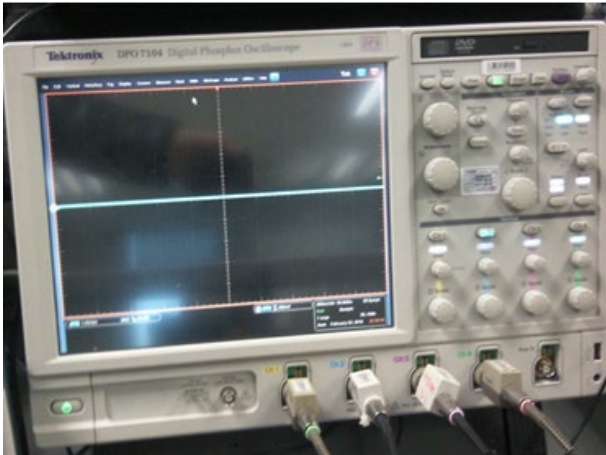


8. Level 3 Repair



8. Level 3 Repair

8-3. Flow chart of Troubleshooting.



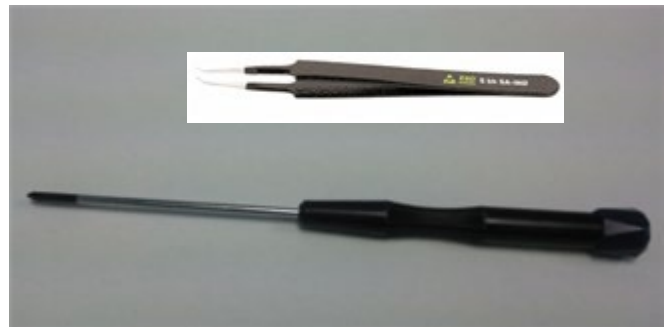
Oscilloscope



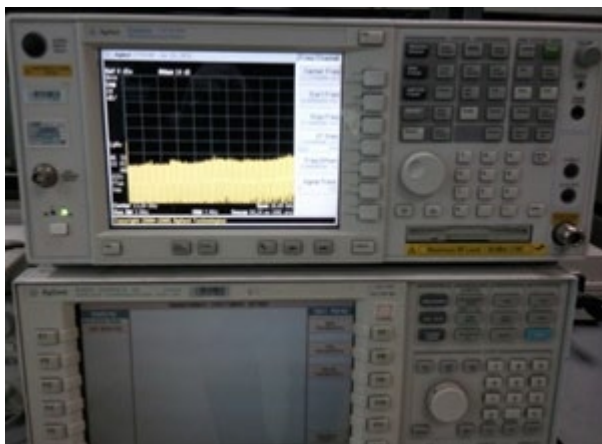
Digital multimeter



Power Supply



+ driver, ESD Safe Tweezer



8960 & Spectrum Analyzer

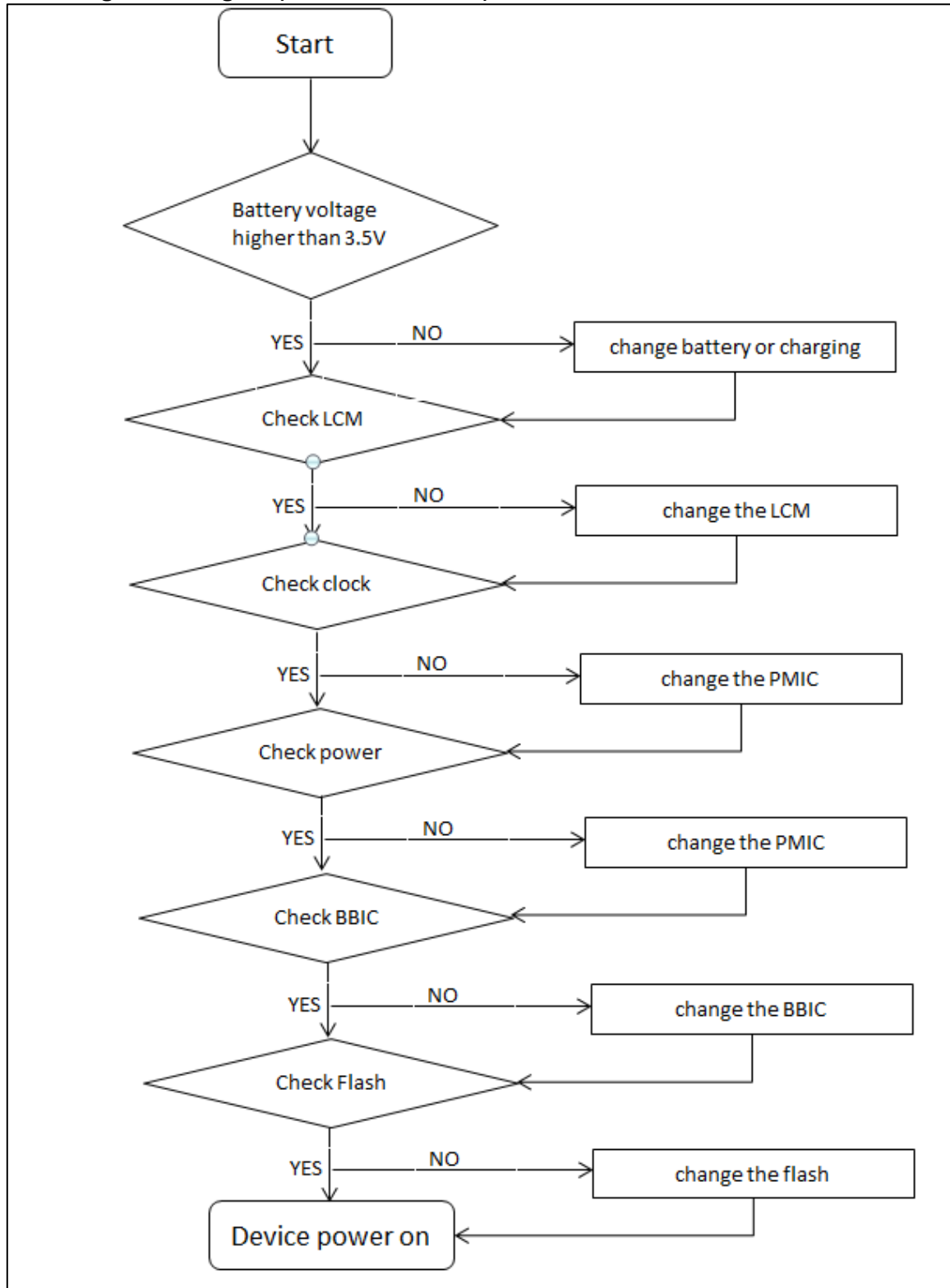


Soldering iron

8. Level 3 Repair

8-4-1. Power On

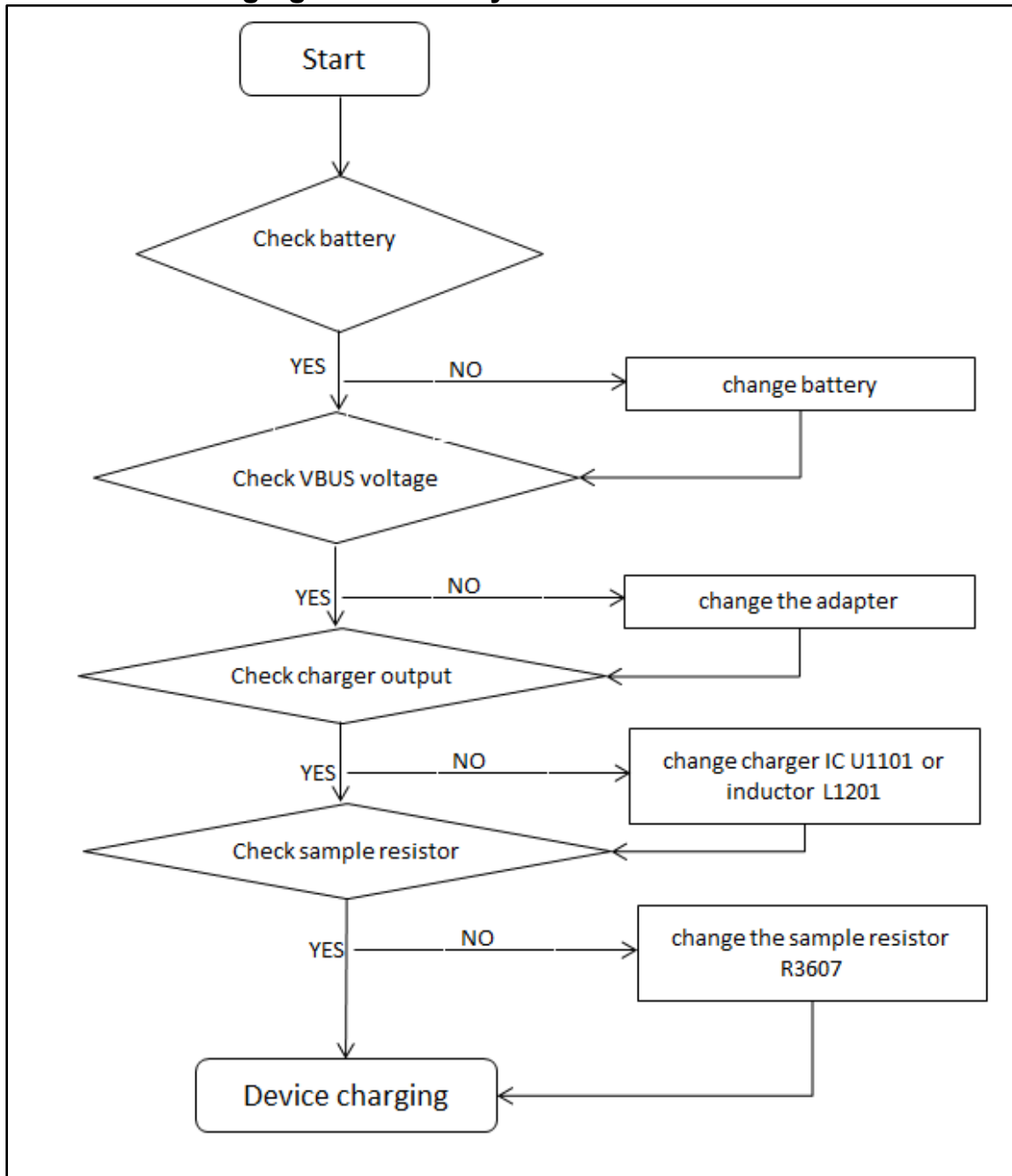
: Checking Power signal (LCM, PMU, Clock)



8. Level 3 Repair

8-4-2. Charging

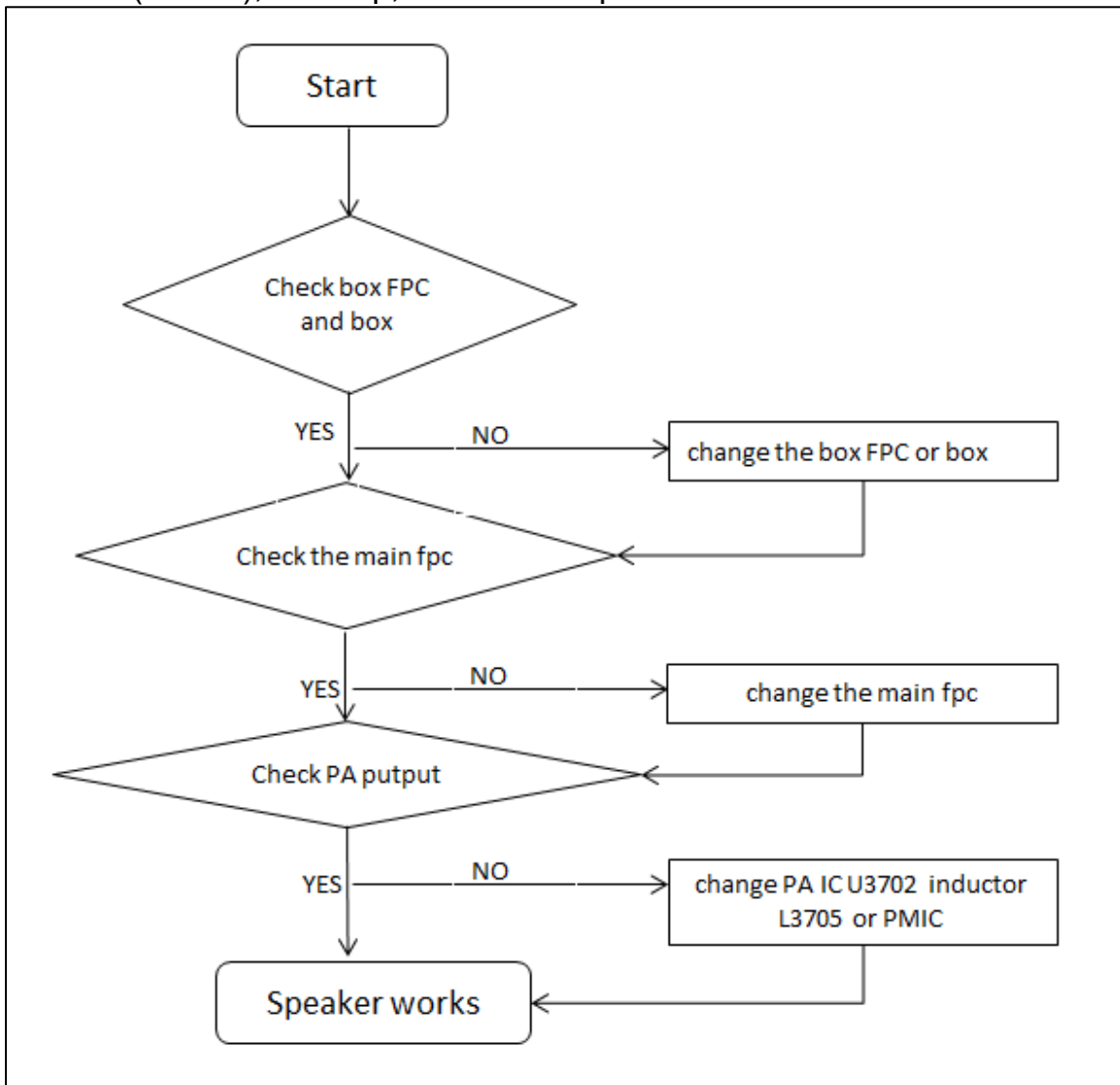
: The charging controlled by the MT6371



8. Level 3 Repair

8-4-3. Audio speaker

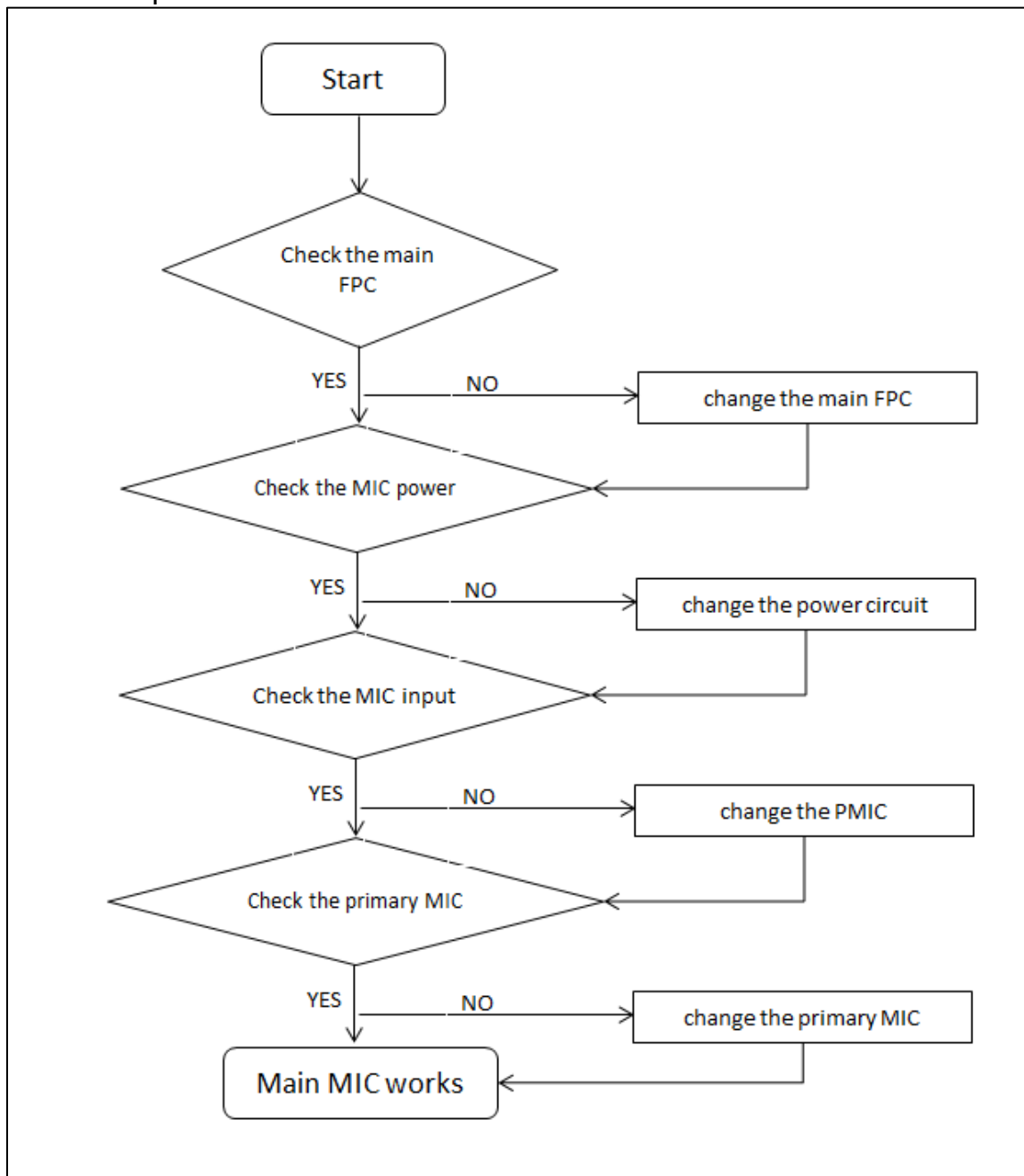
: The Speaker control signals are generated by MT6357 (U701) and PA AW88194 (U3702), the chip, PA and the speaker are to be checked out.



8. Level 3 Repair

8-4-4. Audio Main MIC

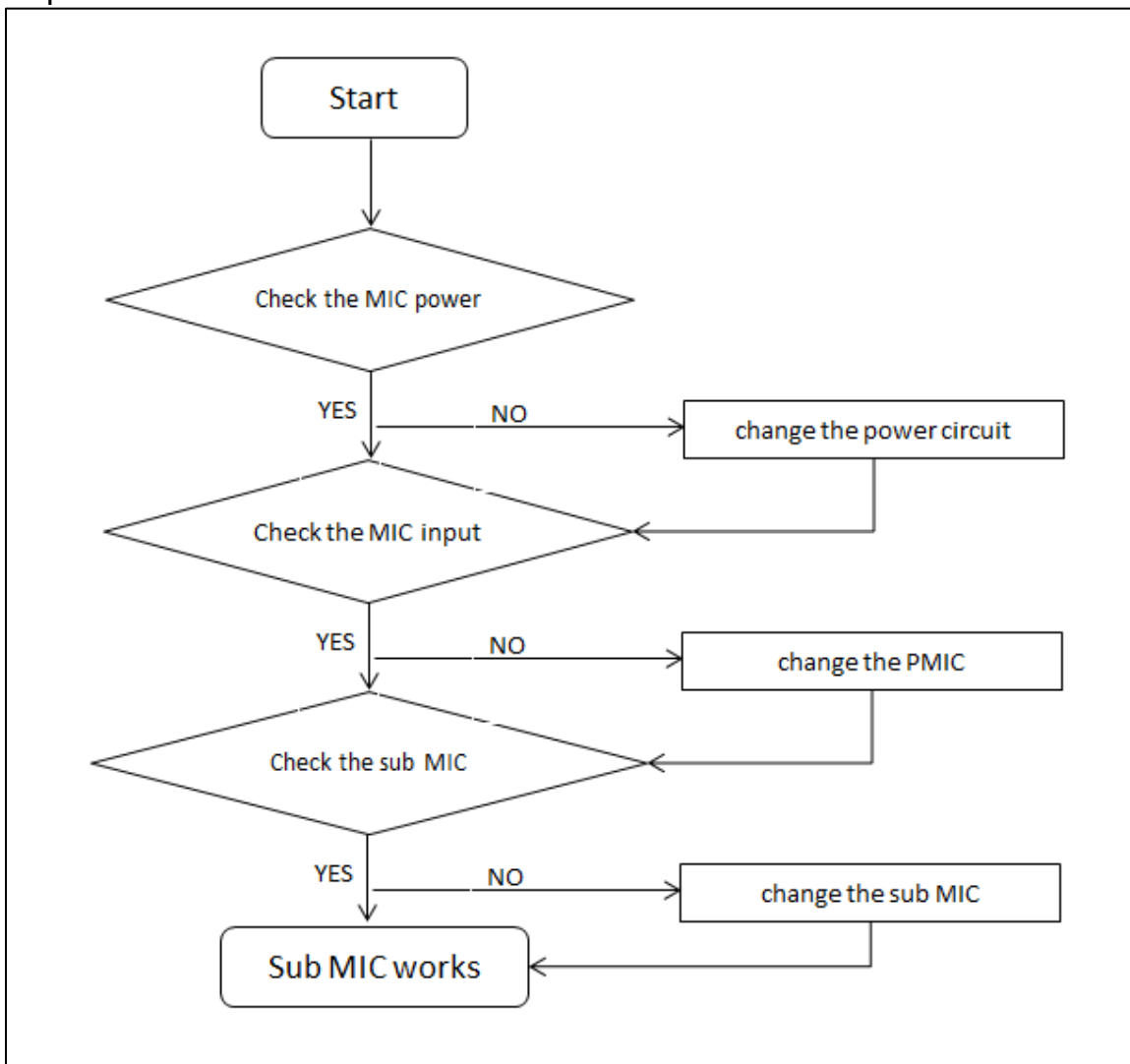
: The MIC control signals are generated by PMU chip MT6357 (U701), the PMU chip and the MIC are to be checked out.



8. Level 3 Repair

8-4-5. Sub MIC

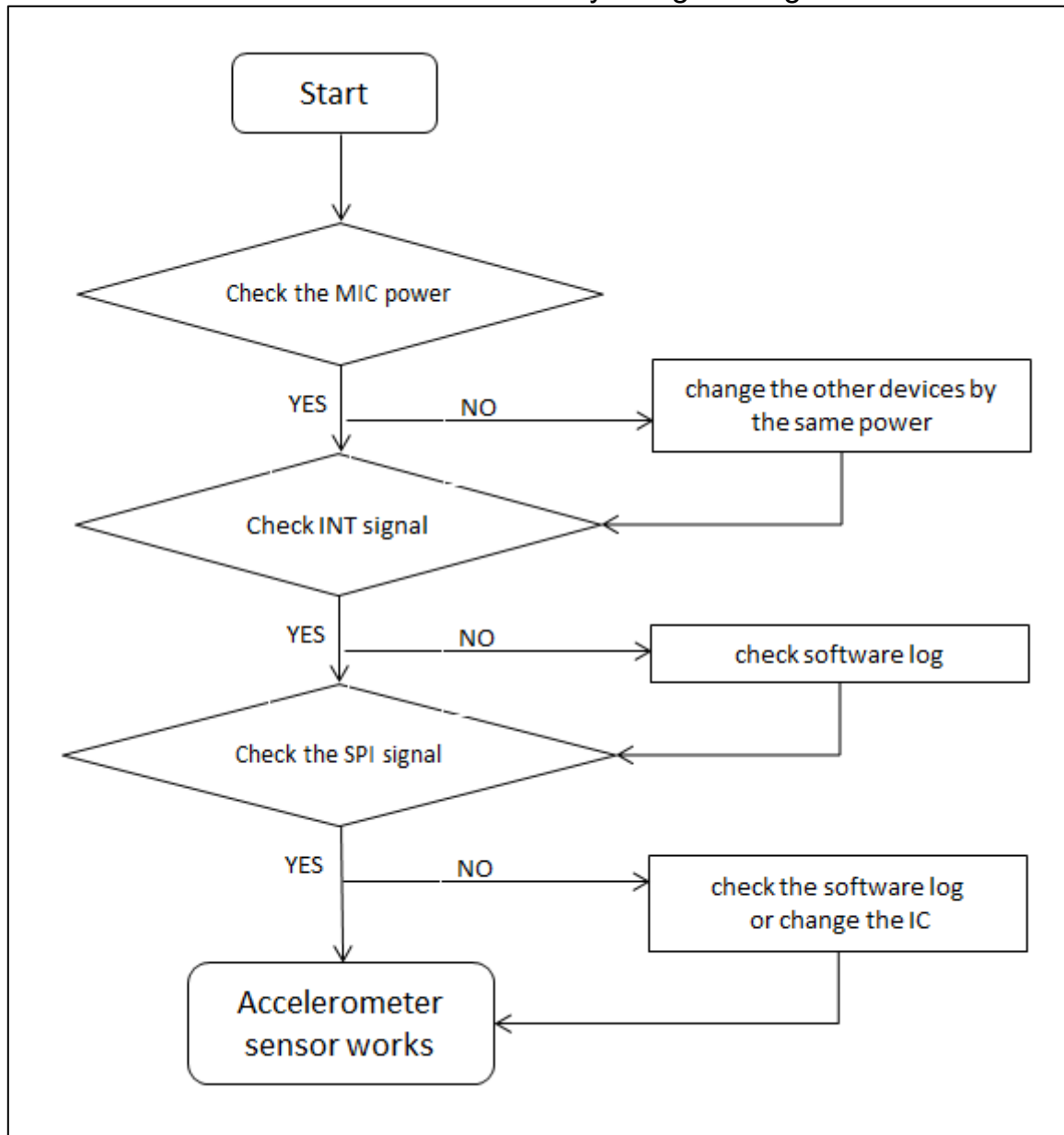
: The MIC control signals are generated by PMU chip MT6357 (U701), the PMU chip and the MIC are to be checked out.



8. Level 3 Repair

8-4-6. Accelerometer sensor

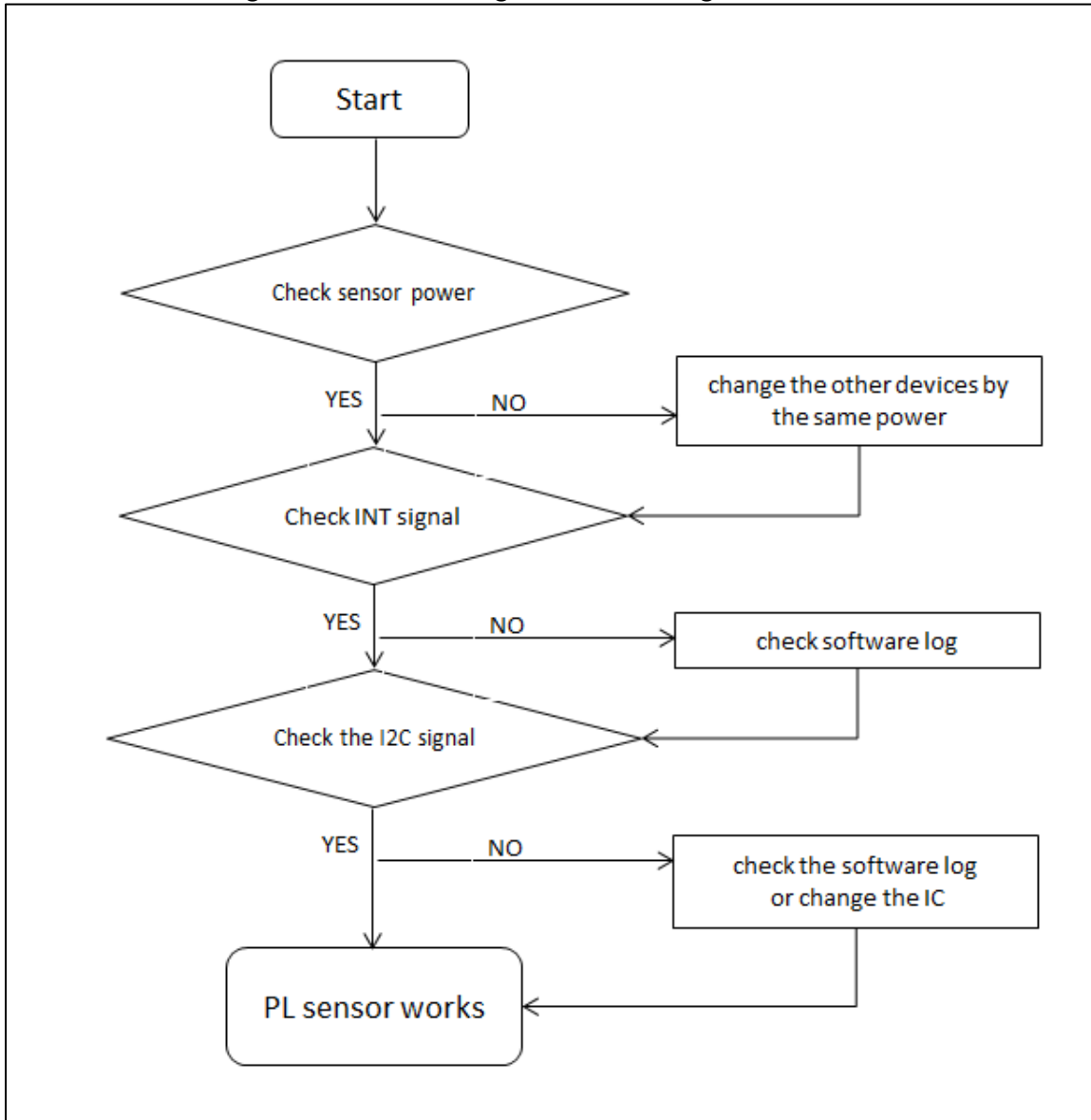
: The Accelerometer sensor is calibrated by using SW algorithm.



8. Level 3 Repair

8-4-7. Proximity and light sensor

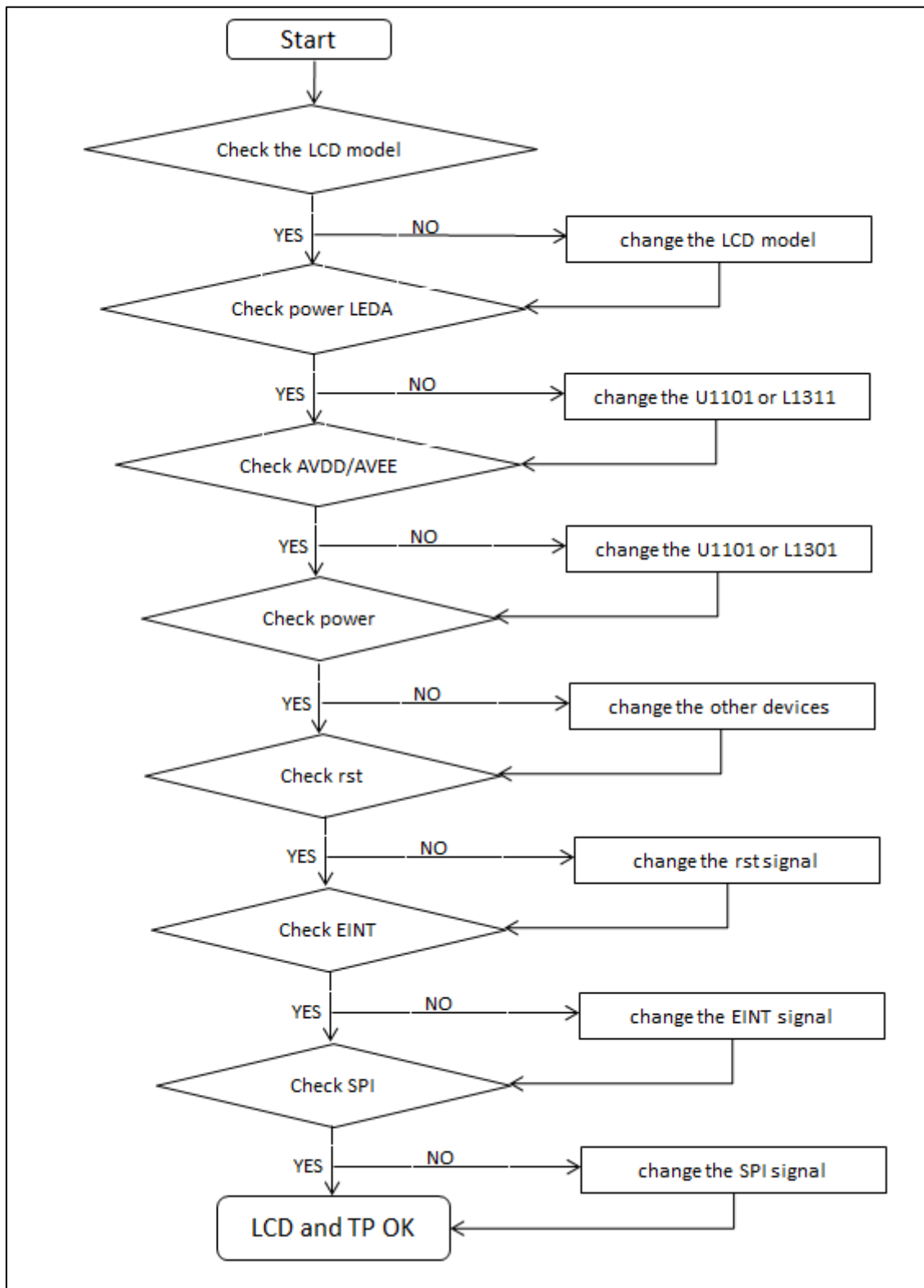
: Proximity (U4417) and Light Sensor (U4416) is worked as below: Control the screen's on/off operation automatically while making phone calls, and adjust the screen brightness according to ambient light.



8. Level 3 Repair

8-4-8. TOUCH SCREEN AND DISPLAY

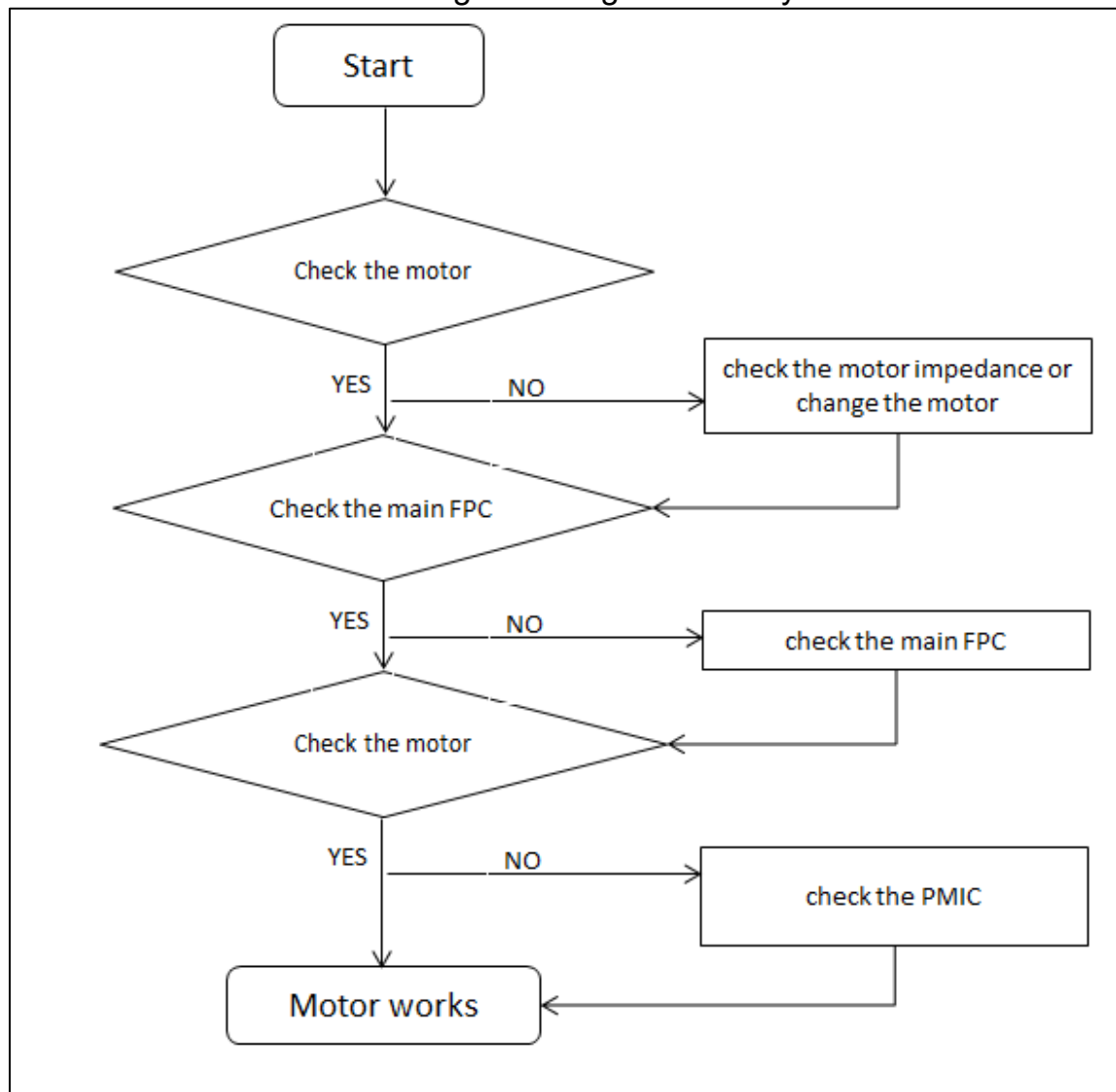
: The Touch control signals are generated by MT6765. It is assembled with LCD.



8. Level 3 Repair

8-4-9. Vibrator

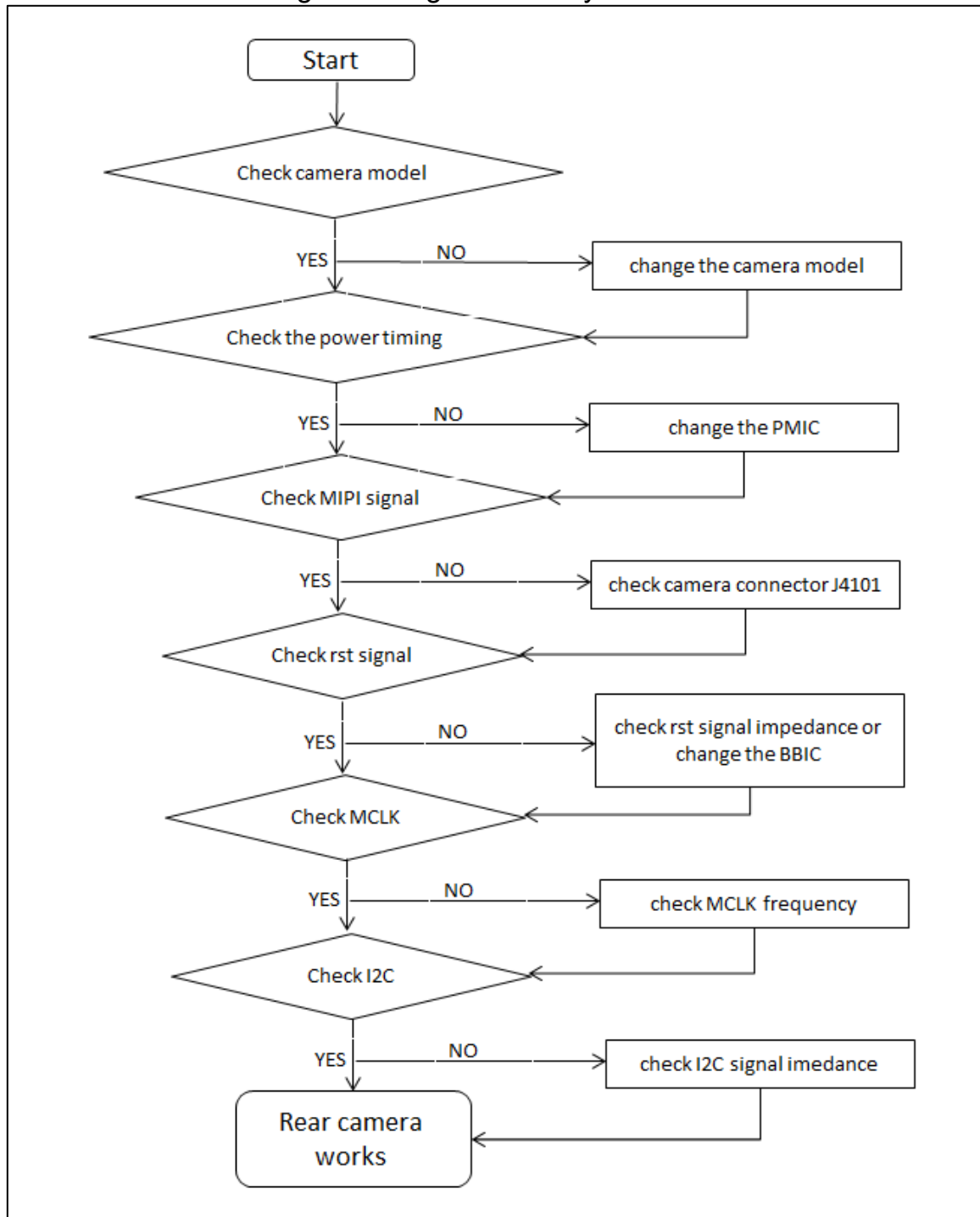
: The Vibrator control signals are generated by MT6357.



8. Level 3 Repair

8-4-10. Rear Camera

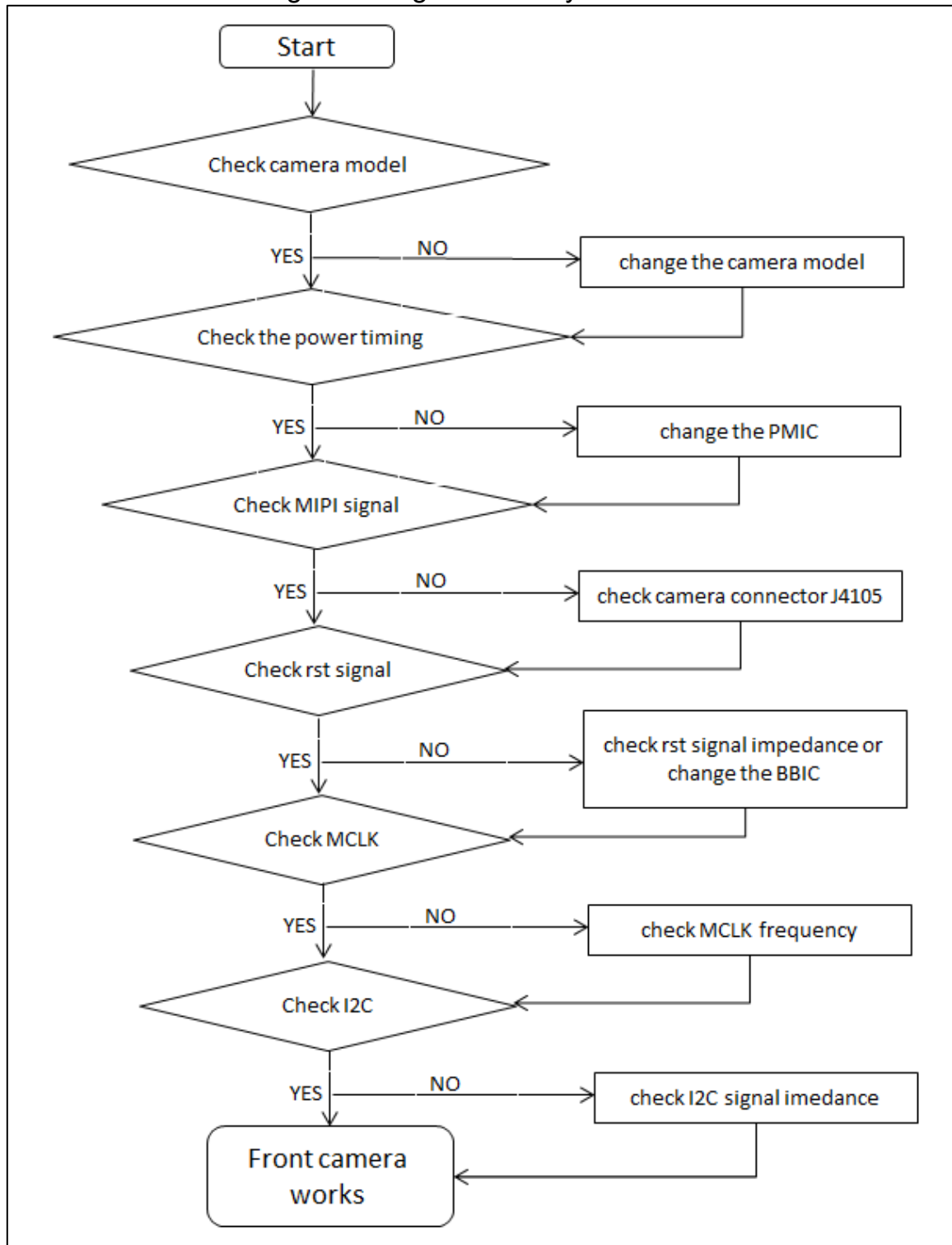
: The Camera control signals are generated by MT6762.



8. Level 3 Repair

8-4-11. Front Camera

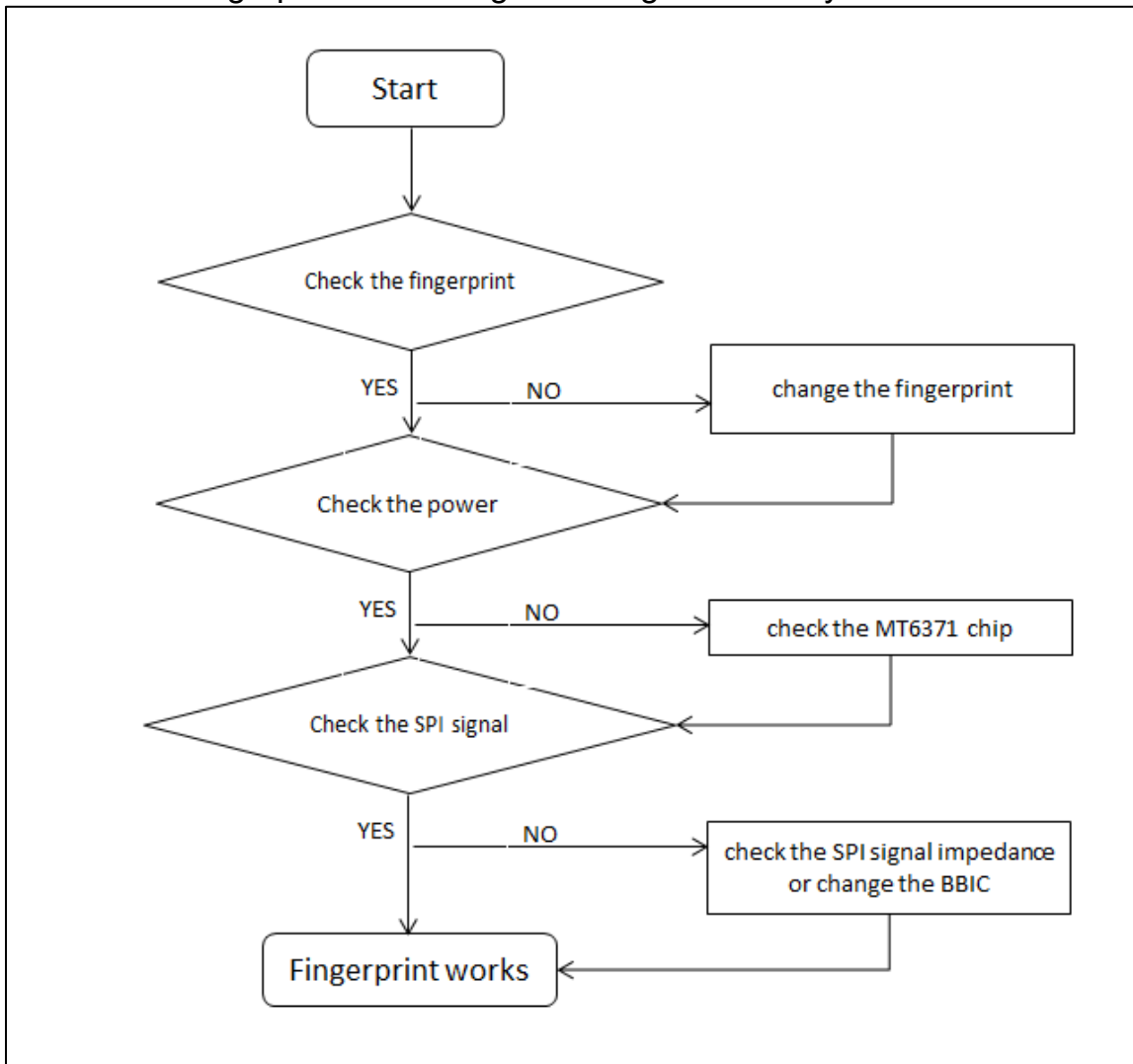
The Camera control signals are generated by MT6765.



8. Level 3 Repair

8-4-12. Fingerprint

The fingerprint control signals are generated by MT6765.



8. Level 3 Repair

8-5. Service Schematics

- U101_MT6765_BB chip IC , Digital Baseband Processor(Top)

558	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	
A	NC	WF_IN	WF_IP	EMIO_CAB		NC		EMIO_CAB			EMIO_DQ0		EMIO_DQ1		EMIO_DQ2		EMIO_DQ3		EMIO_DQ4		EMIO_DQ5		MSDC0_DAT4	MSDC0_DAT2	MSDC0_DAT7	DVDD1_8_MSD	NC	A	
B	WF_OP	WF_QN	EMIO_CAB	EMIO_CAB		EMIO_CAB	EMIO_CAB	NC		EMIO_DQ0	EMIO_DQ1	EMIO_DQ2	EMIO_DQ3	EMIO_DQ4	EMIO_DQ5	EMIO_DQ6	EMIO_DQ7	EMIO_DQ8	EMIO_DQ9	EMIO_DQ10	EMIO_DQ11	EMIO_DQ12	MSDC0_DAT5	MSDC0_DAT6	MSDC0_CMD	MSDC0_DSI	USB_D_M	AVDD1_9_USB	B
C		DVSS	AVDD1_2_VDD	NC	EMIO_CAB	DVSS	EMIO_CAB	NC		EMIO_DQ0	EMIO_DQ1	EMIO_DQ2	EMIO_DQ3	EMIO_DQ4	EMIO_DQ5	EMIO_DQ6	EMIO_DQ7	EMIO_DQ8	EMIO_DQ9	EMIO_DQ10	EMIO_DQ11	EMIO_DQ12	MSDC0_DAT3	MSDC0_RSTB	MSDC0_D_P			C	
D	BT_IN	BT_IP	DVSS	NC	EMIO_CAB	EMIO_CAB	NC	EMIO_CAB	DVSS	EMIO_DQ0	EMIO_DQ1	EMIO_DQ2	EMIO_DQ3	EMIO_DQ4	EMIO_DQ5	EMIO_DQ6	EMIO_DQ7	EMIO_DQ8	EMIO_DQ9	EMIO_DQ10	EMIO_DQ11	EMIO_DQ12	MSDC0_DAT3	MSDC0_CLK	MSDC0_P		SYSRST_B	AVDD1_9_USB	D
E		BT_OP	DVSS	EMIO_CAB	EMIO_CAB	EMIO_CAB	EMIO_CAB	EMIO_CAB	EMIO_CAB	EMIO_DQ0	EMIO_DQ1	EMIO_DQ2	EMIO_DQ3	EMIO_DQ4	EMIO_DQ5	EMIO_DQ6	EMIO_DQ7	EMIO_DQ8	EMIO_DQ9	EMIO_DQ10	EMIO_DQ11	EMIO_DQ12	MSDC0_DAT3	MSDC0_CLK	MSDC0_P		SYSRST_B	AVDD1_9_USB	E
F	GPS_I	CONN_HRST	DVSS	BT_OP	DVSS	DVSS	DVSS	DVSS	EMIO_CAB	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	F
G	GPS_O	CONN_HRST	DVSS	BT_OP	DVSS	DVSS	DVSS	DVSS	EMIO_CAB	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	G
H		AVDD1_8_IOR	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	H
J	DVDD1_8_IOR	CAM_C_LK2	SC16	SDA6	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	J
K		CAM_P_DN2	CAM_R_ST2	ANT_5_E12	ANT_5_E10	ANT_5_E11	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	CONN_TOP_D	K
L	CS1A_L1P	CS1A_L1N	CS1A_L1P	CS1A_L1N	CS1B_L1P	CS1B_L1N	DVSS																						L
M	CS1A_L2P	CS1A_L2N	CS1B_L1P	CS1B_L1N	CS1B_L2P	CS1B_L2N	DVSS																						M
N	CS1A_L1P	CS1A_L1N	CS1A_L2P	CS1A_L2N	CS1B_L1P	CS1B_L1N	DVSS																						N
P	CS1A_L2P	CS1A_L2N	CS1B_L1P	CS1B_L1N	CS1B_L2P	CS1B_L2N	DVSS																						P
R	CS2A_L1P	CS2A_L1N	CS2A_L2P	CS2A_L2N	CS2B_L1P	CS2B_L1N	DVSS																						R
T	CS2A_L1P	CS2A_L1N	CS2A_L2P	CS2A_L2N	CS2B_L1P	CS2B_L1N	DVSS																						T
U	DVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	AVDD1_8_IOR	U
V	CAM_R_ST1	CAM_R_ST2	SDA2	SC12	SDA4																								V
W		CAM_C_LK1	ENT10	ENT9	CAM_C_LK1	CAM_C_LK1	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	ENT11	W
Y	SRCLK_NAI	CAM_P_DN1	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	ENT7	ENT6	Y
AA	KPRO_W0	UTXD0	URXD0	ENT3	ENT2	ENT3	ENT2	ENT3	ENT2	ENT3	ENT2	ENT3	ENT2	ENT3	ENT2	ENT3	ENT2	ENT3	ENT2	ENT3	ENT2	ENT3	ENT2	ENT3	ENT2	ENT3	ENT2	ENT3	AA
AB	KPRO_W1	KPCOL_1	KPCOL_0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	ENT0	AB
AC	SPI0_M_0	SPI0_M_1	SPI0_M_2	SPI0_M_3	SPI0_M_4	SPI0_M_5	SPI0_M_6	SPI0_M_7	SPI0_M_8	SPI0_M_9	SPI0_M_10	SPI0_M_11	SPI0_M_12	SPI0_M_13	SPI0_M_14	SPI0_M_15	SPI0_M_16	SPI0_M_17	SPI0_M_18	SPI0_M_19	SPI0_M_20	SPI0_M_21	SPI0_M_22	SPI0_M_23	SPI0_M_24	SPI0_M_25	SPI0_M_26	SPI0_M_27	AC
AD	SPI0_M_0	SPI0_M_1	SPI0_M_2	SPI0_M_3	SPI0_M_4	SPI0_M_5	SPI0_M_6	SPI0_M_7	SPI0_M_8	SPI0_M_9	SPI0_M_10	SPI0_M_11	SPI0_M_12	SPI0_M_13	SPI0_M_14	SPI0_M_15	SPI0_M_16	SPI0_M_17	SPI0_M_18	SPI0_M_19	SPI0_M_20	SPI0_M_21	SPI0_M_22	SPI0_M_23	SPI0_M_24	SPI0_M_25	SPI0_M_26	SPI0_M_27	AD
AE	SCL1	SPI0_M_0	SPI0_M_1	SPI0_M_2	SPI0_M_3	SPI0_M_4	SPI0_M_5	SPI0_M_6	SPI0_M_7	SPI0_M_8	SPI0_M_9	SPI0_M_10	SPI0_M_11	SPI0_M_12	SPI0_M_13	SPI0_M_14	SPI0_M_15	SPI0_M_16	SPI0_M_17	SPI0_M_18	SPI0_M_19	SPI0_M_20	SPI0_M_21	SPI0_M_22	SPI0_M_23	SPI0_M_24	SPI0_M_25	SPI0_M_26	AE
AF	SDA1	SPI0_M_0	SPI0_M_1	SPI0_M_2	SPI0_M_3	SPI0_M_4	SPI0_M_5	SPI0_M_6	SPI0_M_7	SPI0_M_8	SPI0_M_9	SPI0_M_10	SPI0_M_11	SPI0_M_12	SPI0_M_13	SPI0_M_14	SPI0_M_15	SPI0_M_16	SPI0_M_17	SPI0_M_18	SPI0_M_19	SPI0_M_20	SPI0_M_21	SPI0_M_22	SPI0_M_23	SPI0_M_24	SPI0_M_25	SPI0_M_26	AF
AG	NC	SPI0_M_0	SPI0_M_1	SPI0_M_2	SPI0_M_3	SPI0_M_4	SPI0_M_5	SPI0_M_6	SPI0_M_7	SPI0_M_8	SPI0_M_9	SPI0_M_10	SPI0_M_11	SPI0_M_12	SPI0_M_13	SPI0_M_14	SPI0_M_15	SPI0_M_16	SPI0_M_17	SPI0_M_18	SPI0_M_19	SPI0_M_20	SPI0_M_21	SPI0_M_22	SPI0_M_23	SPI0_M_24	SPI0_M_25	SPI0_M_26	AG
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28		

Figure 2-1. Ball map view